

Getting started

This page demonstrates basic use of the Apparatus. An Apparatus is like a [thread pool](#), but it abstracts away the distribution pattern.

General workflow:

1. Write general Python functions:

```
def f(x):  
    ...
```

2. Create an apparatus and map over data:

```
app = LocalApparatus('experiment-1', num_workers=  
results = app.map(f, data)
```

3. Swap out the apparatus depending on your workload:

```
app = ModalApparatus('experiment-1').w(gpu='A100'  
results = app.map(f, big_data)
```

Each apparatus comes with a *volume* — a shared storage area that persists across function calls. Functions call `get_data_dir()` to read and write files in it, so you can chain steps together: one function prepares data, the next consumes it.

We'll define two functions that pass data through the volume. First, `prep` writes shared configuration; then `train` reads it, runs a mock workload, and saves per-item results.

```
Using Local apparatus "mi-ni-getting-started"  
Wrote config to /home/user/mi-ni/.mini/mi-ni-getting-s  
Results: [10, 20, 30, 40, 50]
```

Volume outputs:

```
--- result_1.txt ---  
input: 1  
result: 10  
lr: 0.001
```

```
--- result_2.txt ---  
input: 2  
result: 20  
lr: 0.001
```

```
--- result_3.txt ---  
input: 3  
result: 30  
lr: 0.001
```

```
--- result_4.txt ---  
input: 4  
result: 40  
lr: 0.001
```

```
--- result_5.txt ---  
input: 5  
result: 50  
lr: 0.001
```